

Technical Reference & Results Sheet: Inverse-FNO-Damage

Mathematical Formulation, Observability Values, and Benchmark Statistics

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Unit Tests Passing

1. Equations of Motion & Observation Model

Transient dynamic equilibrium under seismic ground acceleration $a_g(t)$:
 $M \ddot{u}(t) + C \dot{u}(t) + K(d) u(t) = -M r a_g(t)$, with member stiffness $E_e(d_e) = (1 - d_e)E_0$, $d_e \in [0, 0.5]$.
Observation with 2% sensor noise: $y(t) = H_S [\ddot{u}(t) + r a_g(t), u(t)] + \eta(t)$.
Bilateral difference unit vector: $v_{AB} = (d_A - d_B) / ||d_A - d_B||_2 = 1/\sqrt{2} [1, -1, 0, \dots]^T$. Ground truth separation: $||d_A - d_B||_2 = 0.4243$.

2. Observability & Noise-Whitened Fisher Sensitivity

Linearized sensitivity Jacobian: $J = \partial y / \partial d \in \mathbb{R}^{(C \cdot T) \times E}$. Noise-whitened Jacobian: $J_w = \Sigma_\eta^{-1/2} J$.
Directional Fisher sensitivity along bilateral axis: $\sqrt{(I_{AB})} = ||J_w v_{AB}||_2 = \sqrt{(v_{AB}^T J_w^T J_w v_{AB})}$.

Suite	Sensors	Channels	Fisher $\sqrt{(I_{AB})}$	S0 Ratio	SVD Near-Null Alignment	Physical Status
S0	Horizontal Floor Accelerations	3	5.62	1.0×	$ \langle v_E, v_{AB} \rangle = 0.9982$	Near-null space baseline. Symmetric shear.
S2	Horiz Accel + Column Axial Strain	9	1344.18	239.3×	$ \langle v_E, v_{AB} \rangle = 0.0412$	High Fisher sensitivity; rocking load transfer.
S1	Horiz Accel + Vertical Joint Accel	9	1829.63	325.7×	$ \langle v_E, v_{AB} \rangle = 0.0298$	Joint vertical rotation breaks symmetry.
S4	Multimodal Union (Horiz + Vert + Strain)	15	2270.54	404.2×	$ \langle v_E, v_{AB} \rangle = 0.0185$	Maximal directional sensitivity.

3. Single-Structure Benchmark Performance (Phase 6.2, N = 30 Held-Out Tests)

Suite	Successes	Attribution Acc	Two-Sided p-value	95% Clopper-Pearson CI	Directional Cosine	Separation $ \Delta d_{\hat{}} _2$	True Separation Ratio
S0	15 / 30	50.0%	p = 1.000	[31.3%, 68.7%]	-0.1151 ± 0.122	0.000009	0.002%
S2	15 / 30	50.0%	p = 1.000	[31.3%, 68.7%]	+0.1047 ± 0.111	0.000328	0.08% (Learnability Gap)
S1	24 / 30	80.0%	p = 0.0014	[61.4%, 92.3%]	+0.7272 ± 0.442	0.1576	37.15%
S4	27 / 30	90.0%	p = 9.0 × 10 ⁻⁶	[73.5%, 97.9%]	+0.9420 ± 0.098	0.1671	39.40%

4. Training-Budget Trajectory Across Multi-Structure Generalization (Phase 7, P2 S4)

Evaluated on 530 OpenSees dynamic simulations across 120 PEER earthquakes and 10 structural configurations:

Epoch Budget	In-Distribution Acc	In-Dist Cosine	Interpolation (L2A)	Extrapolation (L2B)	Topological OOD (L3)	Separation $ \Delta d_{\hat{}} _2$
12 Epochs	50.0% ± 0.0%	+0.467 ± 0.037	+0.480 ± 0.038	+0.534 ± 0.055	+0.489 ± 0.012	2.68 × 10 ⁻⁶
25 Epochs	50.0% ± 0.0%	+0.852 ± 0.115	+0.892 ± 0.072	+0.802 ± 0.204	+0.777 ± 0.158	7.68 × 10 ⁻⁶
50 Epochs	50.0% ± 0.0%	+0.897 ± 0.038	+0.896 ± 0.016	+0.824 ± 0.096	+0.837 ± 0.090	1.23 × 10 ⁻⁵
100 Epochs	50.0% ± 0.0%	+0.850 ± 0.205	+0.841 ± 0.216	+0.813 ± 0.204	+0.798 ± 0.199	1.83 × 10 ⁻⁵

5. Key Takeaways & Failure Mode Summary

- 1. Observability $\neq$ Learnability:** High Fisher information ($S_2 = 1344.2$) fails under standard gradient descent because floor accelerations dominate $>98\%$ of the backprop gradient norm over micro-strains.
- 2. Direction–Magnitude Decoupling:** Under cross-structure distribution shift, directional orientation transfers zero-shot ($\cos \approx +0.85$), but finite damage separation collapses ($\sim 10^{-5}$), pinning discrete attribution at 50% chance.
- 3. Optimization Bounds:** Scaling training budget by $8.3\times$ ($12 \rightarrow 100$ epochs) refines directional orientation (up to $+0.996$) but cannot recover finite separation without new representations.